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Title / Título

PROGRAM AGUJERO USER'S MANUAL

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	Revisions Record Registro de Revisiones Pag. 2

Summary / Resumen

The purpose of this document is the definition of the way to use the program AGUJERO.

This program calculates the stress distribution around a hole in an isotropic (metallic) panel under the following load conditions:

- An unloaded hole subjected to by-pass axial load.
- An unloaded hole subjected to by-pass shear load.
- A loaded hole transferring axial load.
- A loaded hole transferring shear load.

Once the stress distributions have been calculated for each one of the load cases specified by the user, all of them are superimposed and then, the stress concentration factor is calculated as well as the point in the contour of the hole where this stress concentration appears.

The load distributions can have any orientations. Because of this, it is possible to study biaxial load states.

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1 INTRODUCTION

1.1 TECHNICAL SHEET OF THE PROGRAM

Technical Report: TEA-NT-2110003

PROGRAM
DATE

AGUJERO

05/2021

OBJECTIVE

This program calculates stress distribution around a hole in a panel under a combination of different load conditions.

LANGUAGE
DEVELOPMENT TOOLS
OS/PLATFORM

Fortran 77

Intel Parallel Studio

Linux and Windows (x64)

Visual Basic Net

MS Visual Studio 2015

Windows (x64)

NUMBER OF LINES (APPROX.)
EXECUTION MODE

2400

Interactive

Notes:

- Blank lines and commented lines excluded
- Count for Visual Basic code excludes lines created by Visual Studio in designer.vb files

AUTHOR
VERSION

Darío Fernández Villalba

1.0 - 3.5

Darío Fernández Villalba / M.Carmen Blanquer

4.0

ADDITIONAL DOCUMENTS

See list of applicable references in chapter §1.3.

1.2 SOFTWARE PURPOSE AND DESCRIPTION

This report makes up the User's Manual for the AGUJERO Program, which has been developed by Airbus Defence and Space from the year 1996 to date.

This program calculates stress distribution around a hole in an isotropic (metallic) panel under the following load conditions:

- An unloaded hole subjected to by-pass axial load.
- An unloaded hole subjected to by-pass shear load.
- A loaded hole transferring axial load.
- A loaded hole transferring shear load.

Once the stress distributions have been calculated for each one of the load cases specified by the user, all them are superimposed and then, the stress concentration factor is calculated as well as the point in the contour of the hole where this stress concentration appears.

The load distributions can have any orientation. By this reason, it is possible to study biaxial load states.

1.3 ABBREVIATIONS AND SYMBOLS

<i>Symbol</i>	<i>Units</i>	<i>Definition</i>
GUI	---	Graphical User Interface
σ_{ref}	MPa	Reference stress
σ_{cir}	MPa	By-pass tension stress
σ_{bg}	MPa	Bearing stress
τ_{cir}	MPa	By-pass shear stress
D	mm	Diameter
Dbl	mm	Distance to the edge measured in the direction of the reference axis X_{ref}
Db	mm	Distance to the edge measured in the direction of the reference axis Y_{ref}
E_p	MPa	Young modulus of the panel
E_r	MPa	Young modulus of the fastener
Kt	---	Stress concentration factor
P	N	Load transferred by the fastener
t	mm	Thickness

1.4 DIFFERENCES AMONG VERSIONS

- Version 1.0:

Initial version developed in Fortran 77 for Digital VAX computer.

- Version 2.0:

Adaptation of the program for the Sun Solaris computer.

- Version 3.0:

The user can choose the language: Spanish or English.

The program is restructured to allow the creation of a dynamic link library DLL for Windows operative system.

- Version 4.0:

The database Agujero.base has been updated to AgujeroNew.base. Only the results of the shear load transfer cases have changed

2 PROGRAM EXECUTION

2.1 WINDOWS/PC VERSIONS

The Visual Basic executable program Agujero.exe (as well as an electronic copy of this manual and the program examples) is located in the folder:

\\multivac\InfoCalculo\Soft\Agujero

To execute the program, the user has to click twice on the Visual Basic executable program.

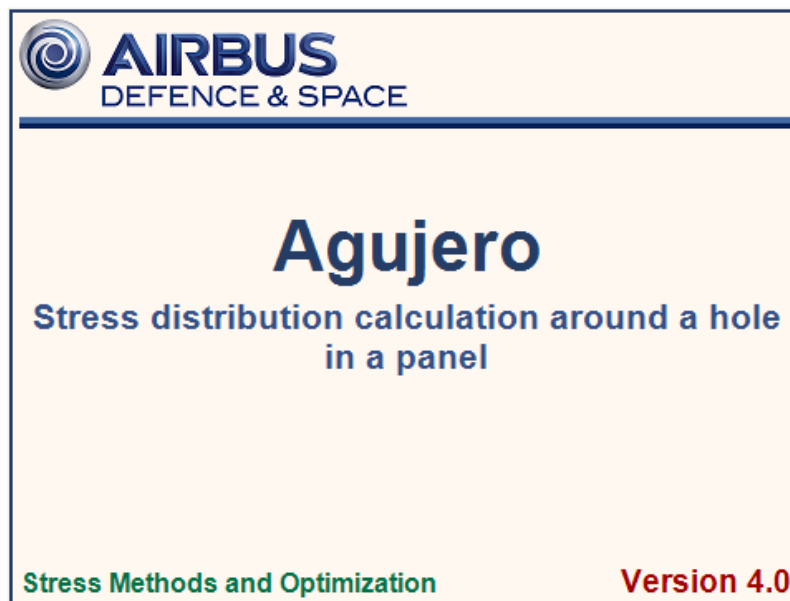


Figure 1. Splash of the Windows GUI version of Agujero

It is also possible to execute the program invoking it by means of a launcher as, for instance, a command file (.bat file). In this case, it is also possible to include some parameters, modifying the behaviour of the program, as described below:

Parameters	Behaviour
Input_File	The program reads the input file and loads all its data into the main form of the interface.
/R Input_File	The program reads the input file, perform the analysis and write the results in the output files without showing any visual interface. The output file will have the same name as the input file, but with extensions .txt.

There is another executable file with the same name, which is the Fortran program in console mode, in the subdirectory .\Consola within that folder (and another copy of the console-mode executable in \\multivac\InfoCalculo\Soft\Consola, this last path than can be added to the Windows system PATH to facilitate its use).

2.2 LINUX/MULTIVAC VERSION

To run interactively this version of the program (console application), you must have already done the assignment of the symbols list in Airbus DS Structural Analysis Office using the command:

```
. asignaciones_calculo
```

This instruction can be executed in the commands terminal prior to execute the program, but it is strongly recommended to include this statement in the files ".profile", and ".kshrc" (if working in Korn Shell) or ".bashrc" (if you work in Bash Shell). To do this, the following lines should be included in the file ".profile":

```
. asignaciones_calculo
ENV=$HOME/.kshrc
export ENV
BASH_ENV=$HOME/.bashrc
export BASH_ENV
```

In files ".kshrc" and ".bashrc" the following line shall be added:

```
. asignaciones_calculo
```

The instruction to run the Linux version of the program in interactive mode in the MULTIVAC3 server is:

```
agujero
```

The program asks for the language to use and the name of the input and output files, as shown in the following example:

```
Idioma / Language:
=====

1 ... ESPAÑOL

2 ... ENGLISH

Idioma / Language: 2

Name of the input file:   Example.dat

Name of the output file: Example.lis
```

The CPU time for calculation depends on the number of problems to solve included in the input file and the number of load cases in each one being, in general, negligible.

The console application can be executed in one single command line, including the input and output filenames and the language as parameters. The above example can be executed as follows:

```
agujero Example.dat Example.lis 2
```

If the third parameter (language) is omitted, the English language will be used as default.

3 INPUT DATA

Below, it is described the structure of the input files, which are required for the correct execution of the program Agujero as well as the Graphic User Interface.

3.1 GRAPHIC USER INTERFACE FOR WINDOWS

The input data can be entered by means of a file or using the graphical interface. If an input file is used, it is necessary to click the button with the open-icon in the input data form of the program and to choose the name of the file.

Load case		Parameters defining the load case			
1	By-pass axial load	100	90	1000	0
2	By-pass axial load	60	0	1000	0
3	By-pass shear load	15	0	1000	0

Figure 2. Input data form

In the input data form of the GUI, as shown above, all needed data to analyse the problem can be directly input.

It is also possible to specify more than one load cases combination by adding more tabs to the form, by clicking on the button «New run». Each configuration can be specified in each tab.

During the input data in the main form, other secondary forms will be deployed in order to do more friendly the process.

The program can be executed clicking the button **Program execution**.

3.2 INPUT DATA FILE

This file contains all of the data necessary to define the problem, as well as the different options required for its execution.

Basically, it contains eight groups of data, although not all of them are strictly necessary. These groups are numbered from one to eight.

The group eight must be the last one, but the others do not need be kept in order. Each set of data groups (up to eight) establishes the procedure for the resolution of one individual problem.

If the user wants to solve several different problems, the input file should contain as many sets of data as the number of problems to be solved. The structure of each group of data is explained below:

Group 1: The title of the problem

Four lines of title (66 characters at most, for each one of them), which will appear at the head of each page in the output file:

- Line 1 -----> 1 First line of title
- Line 2 -----> Second line of title
- Line 3 -----> Third line of title
- Line 4 -----> Fourth line of title

Group 2: General data.

This group consists of two lines containing the following data:

- Line 1 -----> 2 Optional Comment
- Line 2 -----> D t σ_{ref}

Where: D = Diameter of the hole.

t = Thickness of the panel around the hole.

σ_{ref} = Reference stress, selected by the user for calculation of the stress concentration factor Kt. The far field stress is frequently used.

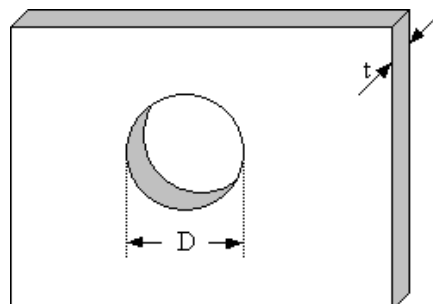


Figure 3. General data.

Group 3: Cases of by-pass axial load.

The structure of the data in this group is the following:

Line 1	----->	3	Optional comment			
Line 2	----->	N_1	$\sigma_{cir}(1)$	$\theta(1)$	$Db(1)$	$Er/Ep(1)$
Line 3	----->		$\sigma_{cir}(2)$	$\theta(2)$	$Db(2)$	$Er/Ep(2)$
						
						
Line N_1+1	----->		$\sigma_{cir}(N_1)$	$\theta(N_1)$	$Db(N_1)$	$Er/Ep(N_1)$

Where: N_1 = Number of cases of by-pass axial load to be considered.

$\sigma_{cir}(i)$ = By-pass axial stress in the case of load "i", positive if it is a tension load.

$\theta(i)$ = Angle in degrees between the X-axis of reference for the case "i" X_{ref} and the absolute X-axis, measured from the absolute X-axis, positive in a counter-clockwise direction. The reference axes are defined so that the reference axis Y_{ref} is in the direction of the axial load, in either of the two senses.

$Db(i)$ = Distance to the closest edge, measured in a perpendicular direction to the load axis of the case "i".

$Er/Ep(i)$ = Ratio between the Young's modulus of the fastener and of the panel for calculating the hole filling effect in case "i". To avoid considering this effect, it is necessary to define this data as zero.

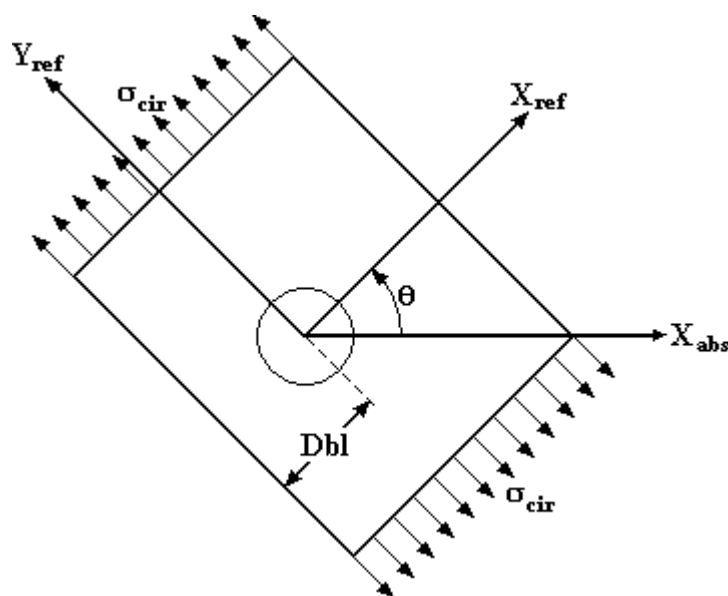


Figure 4. By-pass axial load case.

Group 4: Cases of by-pass shear load.

The structure of data for this group is the following:

Line 1	----->	4	Optional comment			
Line 2	----->	N_2	$\tau_{cir}(1)$	$\theta(1)$	$Db(1)$	$Er/Ep(1)$
Line 3	----->		$\tau_{cir}(2)$	$\theta(2)$	$Db(2)$	$Er/Ep(2)$
						
						
Line N_2+1	----->		$\tau_{cir}(N_2)$	$\theta(N_2)$	$Db(N_2)$	$Er/Ep(N_2)$

Where N_2 = Number of cases of by-pass shear load to be considered.

$\tau_{cir}(i)$ = By-pass shear stress in the case of load "i", positive is tension is produced in the first quadrant of the reference system (see figure below).

$\theta(i)$ = Angle in degrees forming the X-axis of reference for the case "i" X_{ref} , with the absolute X-axis, measured from the absolute X-axis, positive in a counter-clockwise direction. The reference axes are defined so that the reference axis X_{ref} points in the opposite sense of the closest edge.

$Db(i)$ = Distance to the closest edge measured in the direction of the reference axis X_{ref} in case "i".

$Er/Ep(i)$ = Ratio between the Young's modulus of the fastener and of the panel for calculating the hole filling effect in case "i". To avoid considering this effect, it is necessary to define this data as zero.

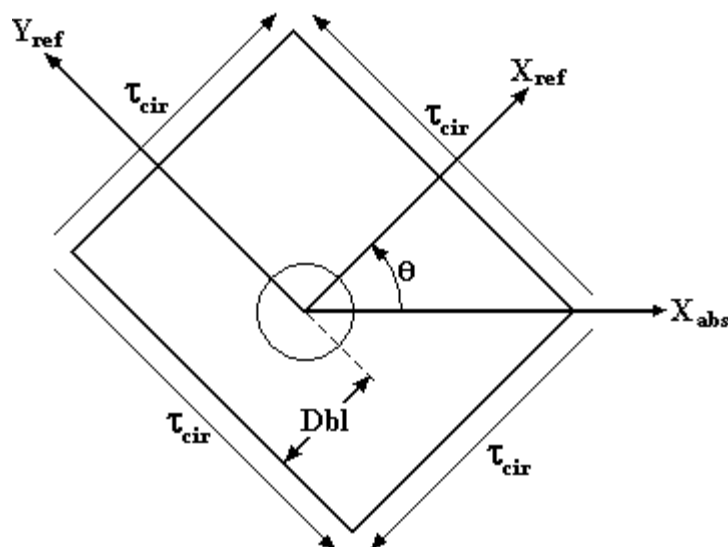


Figure 5. By-pass shear load case.

Group 5: Cases of axial load transfer.

The structure of the data of this group is the following:

Line 1	----->	5	Optional comment			
Line 2	----->	N_3	$P_{ax}(1)$	$\theta(1)$	$Dbl(1)$	$Dbs(1)$
Line 3	----->		$P_{ax}(2)$	$\theta(2)$	$Dbl(2)$	$Dbs(2)$
						
						
Line N_3+1	----->		$P_{ax}(N_3)$	$\theta(N_3)$	$Dbl(N_3)$	$Dbs(N_3)$

Where N_3 = Number of cases of axial load transfer to be considered.

$P_{ax}(i)$ = Load transferred by the fastener in the case "i". It is always positive.

$\theta(i)$ = Angle in degrees forming the X-axis of reference for the case "i" X_{ref} , with the absolute X-axis, measured from the absolute X-axis, positive in a counter-clockwise direction. The reference axes are defined so that the axis Y_{ref} has the same direction and sense as the load transferred by the fastener.

$Dbl(i)$ = Distance to the closest edge measured in the direction of the reference axis, X_{ref} or $-X_{ref}$, in case "i".

$Dbs(i)$ = Distance to the edge measured in the direction and sense of the reference axis Y_{ref} in case "i".

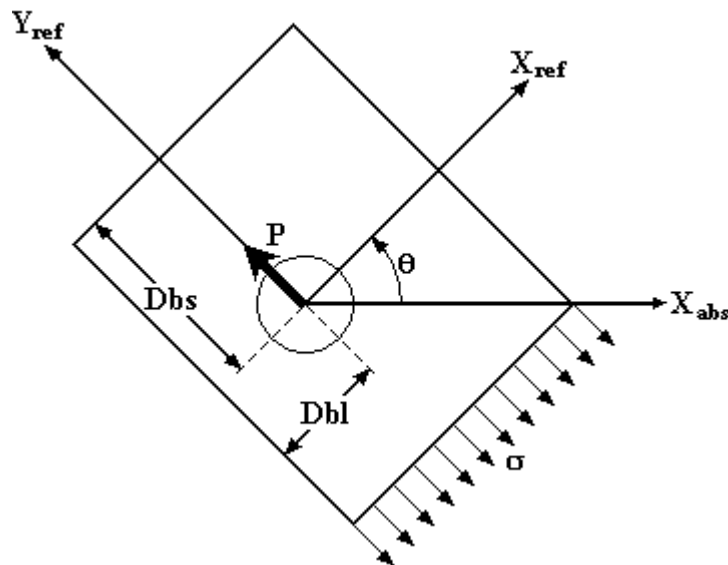


Figure 6. Axial load transfer case.

Group 6: Cases of shear load transfer.

The structure of data of this group is the following:

Line 1	----->	6	Optional comment
Line 2	----->	N_4	$P_{sh}(1)$ $\theta(1)$ $Dbl(1)$ $Pitch(1)$
Line 3	----->		$P_{sh}(2)$ $\theta(2)$ $Dbl(2)$ $Pitch(2)$
			
			
Line N_4+1	----->		$P_{sh}(N_4)$ $\theta(N_4)$ $Dbl(N_4)$ $Pitch(N_4)$

Where N_4 = Number of cases of by-pass shear to be considered.

$P_{sh}(i)$ = Load transferred by the fastener in the case "i". It is positive if it points in the same direction as the positive reference axis Y_{ref} .

$\theta(i)$ = Angle in degrees forming the X-axis of reference for the case "i" X_{ref} , with the absolute X-axis, measured from the absolute X-axis, positive in a counter-clockwise direction. The reference axes are defined so that the reference axis Y_{ref} has the same direction as the load transferred by the fastener and the reference axis X_{ref} points to the furthest edge.

$Dbl(i)$ = Distance to the closest edge measured in the negative direction of the reference axis $-X_{ref}$, in case "i".

$Pitch(i)$ = Average spacing between fasteners (considering only the two adjacent ones), measured in the fastener load direction in case "i".

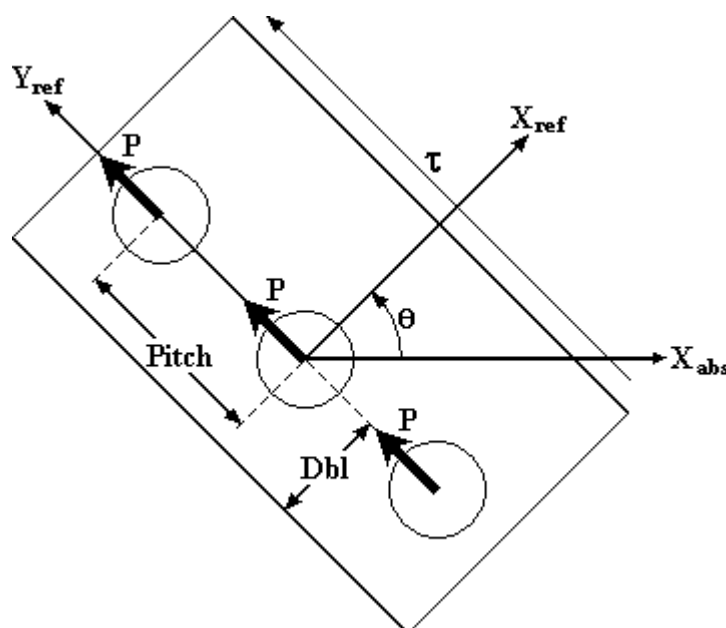


Figure 7. Shear load transfer case.

Group 7: Print control.

The structure of the data of this group is the following:

Line 1 -----> 7 Optional comment

Line 2 -----> I_1 I_2

Where I_1 = The printing code of the stress distributions around the hole for each and every one of the load cases defined in the input data.

I_2 = The printing code of the stress distributions around the hole for the superposition of all load cases defined in the input data.

These codes are defined so that:

$I_i = 0$: The corresponding results are not written in the output file.

$I_i \neq 0$: The corresponding results are written in the output file.

Group 8: Execution of the program

In this group, the execution of the program is invoked and it should appear as often as the problems to be resolved.

The structure of the data in this group is the following:

Line 1 -----> 8 Optional comment

For the first problem (first execution) defined in the input file, the groups number 2, 7 and 8 are always required. Additionally, it is also necessary to include at least one of the groups 3, 4, 5 and 6.

For the rest of problems in the input file, only the group 8 is required. This is because missing groups are assumed to be equal than for the previous problem. For this reason, when a set of load cases has been previously included (group 3, 4, 5 or 6) and it have to be removed, the corresponding group must be included in the input file with zero load cases, as shown below:

- To remove by-pass axial load cases:

```
3 By-pass axial load cases
0
```

- To remove by-pass shear load cases:

```
4 By-pass shear load cases
0
```

- To remove axial load transfer cases:

```
5 Axial load transfer
0
```

- To remove shear load transfer cases:

```
6 Shear load transfer
0
```

4 OUTPUT RESULTS

4.1 WINDOWS PC/GUI

In the Windows/PC GUI version, when all data are entered and the button 'Program execution' has been clicked, the program runs showing a tracking form and saves the results into the output file.

Calculating stress distributions
Writing analysis results
=====

Processing run 3
Reading input data
Writing output data
Calculating stress distributions
Writing analysis results
=====

Processing run 4
Reading input data
Writing output data
Calculating stress distributions
Writing analysis results
=====

End of program

Run 1	Max. Stress	258.4	Kt(Ref)	2.4607
Graph	Angle	108.28	Kt(Brg)	0.0000
Angle			Stress	

Return to data panel Exit from the program

Figure 8. Tracking and results form

4.2 OUTPUT FILE

The AGUJERO program creates a single output file with the analysis results. This file includes the following information:

- Summary of the input data:
 - General geometric data:
 - = Diameter of the holeD
 - = Thickness of the panelt
 - = Reference stress σ_{ref}
 - Definition of the simple load cases:
 - = Cases of by-pass axial load
 - = Cases of by-pass shear load
 - = Cases of transfer of the axial load
 - = Cases of transfer of the shear load

- Stress distributions around the hole:
 - Stress distributions around the hole, every five degrees, for each one of the simple load cases specified in the input.
 - Stress distribution around the hole, every five degrees, for the superimposing of all the simple cases of load.
- Global results:
 - Resultant load applied in the holeP
 - Orientation of the resultant load $\theta(P)$
 - Maximum stress on the hole contour $\sigma_{\max}$
 - Point of the contour in which the stress is the maximum $\theta(\sigma_{\max})$
 - Reference stress..... σ_{ref}
 - Bearing stress..... $\sigma_{\text{bg}} = P / (D \cdot t)$
 - Stress concentration factor referred to the reference stress..... $K_{t_{\text{ref}}} = \sigma_{\max} / \sigma_{\text{ref}}$
 - Stress concentration factor referred to the reference stress..... $K_{t_{\text{bg}}} = \sigma_{\max} / \sigma_{\text{bg}}$

An example of the output file is provided in chapter 5.

When the program detects an error in the input data, it writes the corresponding message in the console and stops the execution.

The error messages that might appear in the console, are listed on next pages:

4.3 WARNING AND ERROR MESSAGES

Error messages. English output:

ERROR: The input file cannot be opened
ERROR: The file with the basic distributions database cannot be opened
ERROR: The file with the basic distributions database contains erroneous data
ERROR: End of file while reading the basic distributions database
ERROR: Erroneous input data in input file
ERROR: Group 1 missing in input file (Title)
ERROR: End of file while reading group 1 in input file (Title)
ERROR: Group 2 missing in input file (General data)
ERROR: Group 2 is defined badly (General data)
ERROR: End of file while reading group 2 in input file (General data)
ERROR: Groups 3, 4, 5 and 6 missing in input file (Load cases)
ERROR: Group 3 is defined badly (Cases of by-pass axial load)
ERROR: End of file while reading group 3 in input file (Cases of by-pass axial load)
ERROR: Group 4 is defined badly (Cases of by-pass shear load)
ERROR: End of file while reading group 4 in input file (Cases of by-pass shear load)
ERROR: Group 5 is defined badly (Cases of axial load transfer)
ERROR: End of file while reading group 5 in input file (Cases axial load transfer)
ERROR: Group 6 is defined badly (Cases of shear load transfer)
ERROR: End of file while reading group 6 in input file (Cases shear load transfer)
ERROR: Group 7 missing in input file (Output results)
ERROR: Group 7 is defined badly (Output results)
ERROR: End of file while reading group 7 in input file (Output results)
ERROR: The output file cannot be opened
ERROR: Too many cases of by-pass axial load, exceeding the program capability
ERROR: Too many cases of by-pass shear load, exceeding the program capability
ERROR: Too many cases of axial load transfer, exceeding the program capability
ERROR: Too many cases of shear load transfer, exceeding the program capability
ERROR: The hole diameter must be positive
ERROR: The panel thickness must be positive
ERROR: The reference stress cannot be zero
ERROR: The distance to the edge must be greater than the radius
ERROR: The Young modulus ratio cannot be negative
ERROR: The transferred load through the fastener must be positive
ERROR: The distance to the end must be greater than the radius
ERROR: The fasteners pitch must be greater than the diameter

Error messages. Spanish output:

ERROR: No se puede abrir el fichero de entrada

ERROR: No se puede abrir el fichero con la base de datos de distribuciones básicas

ERROR: El fichero con la base de datos de distribuciones básicas contiene datos erróneos

ERROR: Fin de fichero mientras se leía la base de datos de distribuciones básicas

ERROR: Datos erróneos en el fichero de entrada

ERROR: Grupo 1 ausente en el fichero de entrada (Título)

ERROR: Fin de fichero mientras se leía el grupo 1 en el fichero de entrada (Título)

ERROR: Grupo 2 ausente en el fichero de entrada (Datos generales)

ERROR: Grupo 2 mal definido (Datos generales)

ERROR: Fin de fichero mientras se leía el grupo 2 en el fichero de entrada (Datos generales)

ERROR: Grupos 3, 4, 5 y 6 ausentes en el fichero de entrada (Casos de carga)

ERROR: Grupo 3 mal definido (Casos de carga axial circulante)

ERROR: Fin de fichero mientras se leía el grupo 3 en el fichero de entrada (Casos de carga axial circulante)

ERROR: Grupo 4 mal definido (Casos de cortadura circulante)

ERROR: Fin de fichero mientras se leía el grupo 4 en el fichero de entrada (Casos de cortadura circulante)

ERROR: Grupo 5 mal definido (Casos de transferencia de carga axial)

ERROR: Fin de fichero mientras se leía el grupo 5 en el fichero de entrada (Casos de transferencia de carga axial)

ERROR: Grupo 6 mal definido (Casos de transferencia de cortadura)

ERROR: Fin de fichero mientras se leía el grupo 6 en el fichero de entrada (Casos de transferencia de cortadura)

ERROR: Grupo 7 ausente en el fichero de entrada (Salida de resultados)

ERROR: Grupo 7 mal definido (Salida de resultados)

ERROR: Fin de fichero mientras se leía el grupo 7 en el fichero de entrada (Salida de resultados)

ERROR: No se puede abrir el fichero de salida

ERROR: Demasiados casos de carga axial circulante, excediendo la capacidad del programa

ERROR: Demasiados casos de cortadura circulante, excediendo la capacidad del programa

ERROR: Demasiados casos de transferencia de carga axial, excediendo la capacidad del programa

ERROR: Demasiados casos de transferencia de cortadura, excediendo la capacidad del programa

ERROR: El diámetro del agujero debe ser positivo

ERROR: El espesor del panel debe ser positivo

ERROR: El esfuerzo de referencia no puede ser cero

ERROR: La distancia al borde debe ser mayor que el radio

ERROR: La relación de módulos de elasticidad no puede ser negativa

ERROR: La carga transferida por el remache debe ser positiva

ERROR: La distancia al extremo debe ser mayor que el radio

ERROR: El paso entre remaches debe ser mayor que el diámetro

5 LIMITATIONS

The main limitations of the program are related to:

Filling effect

The filling effect has been obtained for non-loaded holes but not for holes with load transfer, for which the stress concentration factor due to the hole filling is expected to be lower. Therefore, not considering the filling effect is recommended in order to be conservative ($E_r/E_p = 0$). Its consideration can be a recovery solution in early findings appeared in test or in service.

Superposition principle

- By-pass load cases:

If there are several unitary by-pass load cases but one of them is the prevailing load case, the hole filling effect could be considered but only for the prevailing one.

- Load transfer cases:
 - Since the FE models used to calculate the stress concentration factor include non-linear contacts, the superposition principle is only valid for the combination of cases in which the load has the same direction and sign.
 - The hole filling effect should not be considered when combining load transfer cases with by-pass load cases.

6 EXAMPLES

As an example of how to use the program AGUJERO, four different problems are going to be solved with only one program execution.

Problem 1:

This problem consists in an unloaded hole in an infinite panel subjected to a general stress state without loads out of the panel plane.

The data corresponding to this problem are the following:

- Geometry:
 - Hole diameter: 5 mm.
 - Panel thickness: 2 mm.
- Stress state:
 - Normal – X..... $\sigma_x = 100 \text{ N/mm}^2$
 - Normal – Y..... $\sigma_y = 60 \text{ N/mm}^2$
 - Shear – XY $\tau_{xy} = 15 \text{ N/mm}^2$

The maximum principal stress will be taken as reference stress, that is $\sigma_{\text{ref}} = 105 \text{ N/mm}^2$.

The load state as well as the reference axes in which the stress state is defined can be better understood in the following figure:

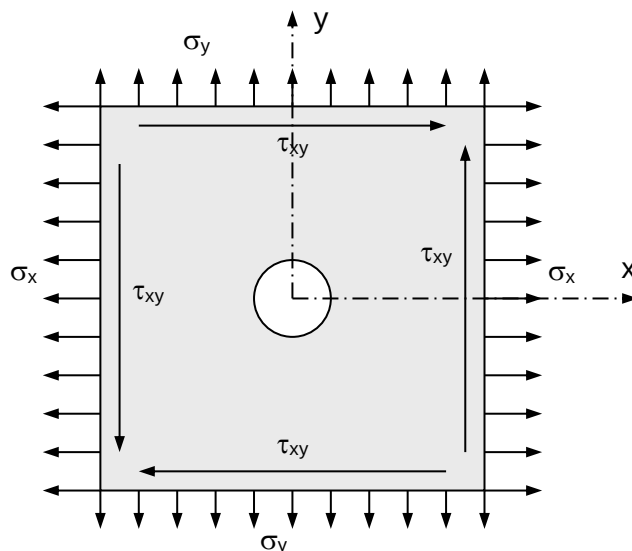


Figure 9. Problem 1 – load state.

The stress state defined in the previous axes will be used, so that the problem can be split in three simple cases, in the following way:

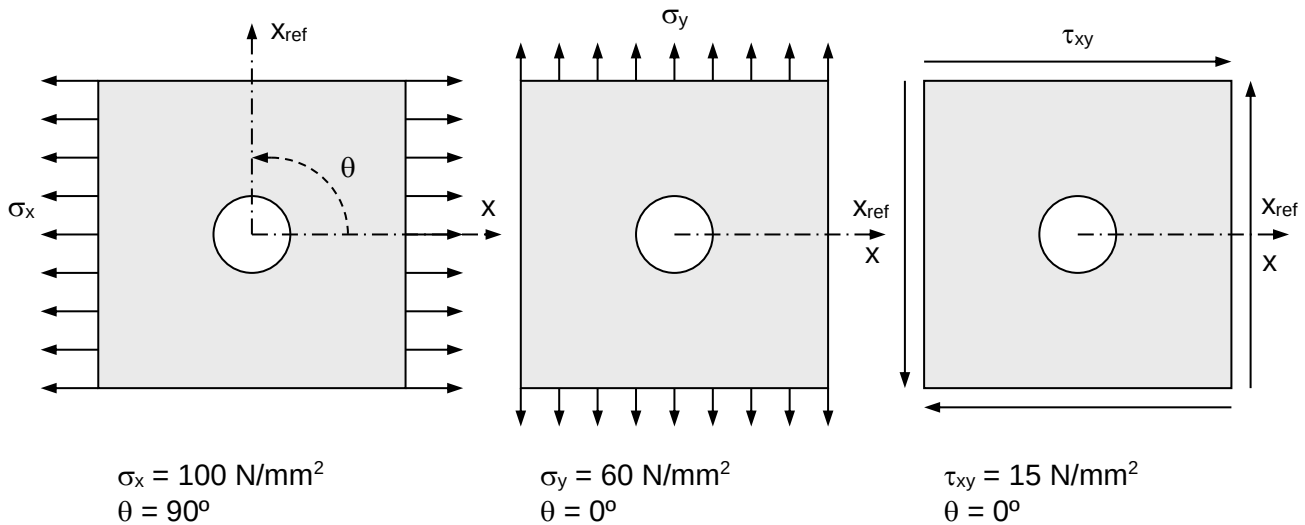


Figure 10. Problem 1 – basic cases.

For this problem, a full output of the results will be requested, that is, stress distributions around the hole for each one of the basic load cases, stress distribution for the resultant case and stress concentration factor.

Problem 2:

The problem is the same as above, but the stress state will be defined in principal axes. These axes form an angle of 18.435 degrees with the previous ones, being the principal stresses 105 and 55 N/mm². This problem, then, has to be split in two basic cases, as follows:

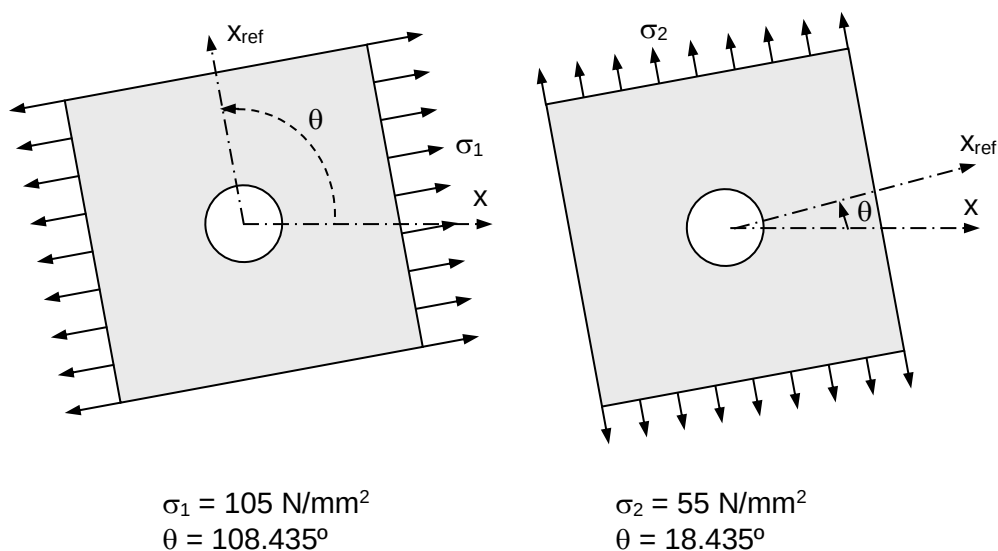


Figure 11. Problem 2 – basic cases.

For this problem, the stress distribution around the hole will only be requested for the resultant load case.

Problem 3:

This problem consists in two plates joined by a fastener and submitted to tensile load, according to the next figure:

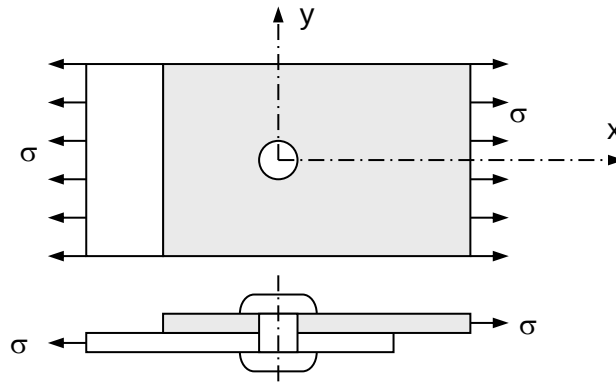
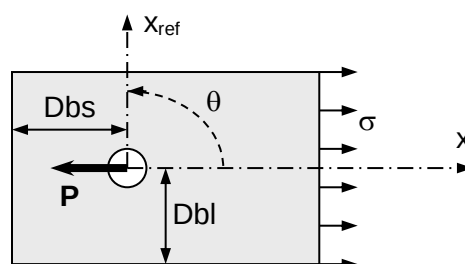


Figure 12. Problem 3 – Load state.

The data corresponding to this problem are the following:

- Geometry:
 - Hole diameter:5 mm.
 - Panel thickness:2 mm.
 - Panel width:20 mm.
 - Distance from hole to unloaded end:.....17.5 mm.
- Stress state:
 - Normal stress: $\sigma = 100 \text{ N/mm}^2$.

This case can be associated to a single basic case, as follows, where the load transferred through the fastener can be calculated by applying the equilibrium of forces:



$$\begin{aligned} \sigma &= 100 \text{ N/mm}^2 & \theta &= 90^\circ \\ Dbl &= 10 \text{ mm} & Dbs &= 17.5 \text{ mm} \\ P &= 2 \cdot \sigma \cdot Dbl \cdot t = 4000 \text{ N} \end{aligned}$$

Figure 13. Problem 3 – basic case.

Since there is only one basic case, only the stress distribution around the hole for the resultant case will be requested.

For Kt calculation purposes, the reaction stress in the far field will be taken as reference stress, therefore $\sigma_{\text{ref}} = 100 \text{ N/mm}^2$.

Problem 4:

This problem consists in two plates joined by a row of fasteners and submitted to shear load, so that all fasteners are transferring the same load, according to the next figure:

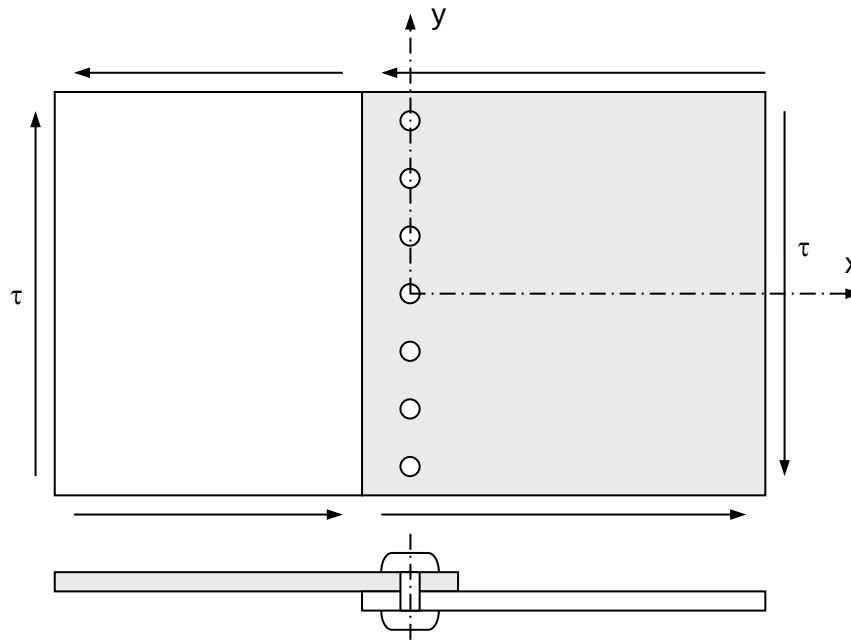


Figure 14. Problem 4 – Load state.

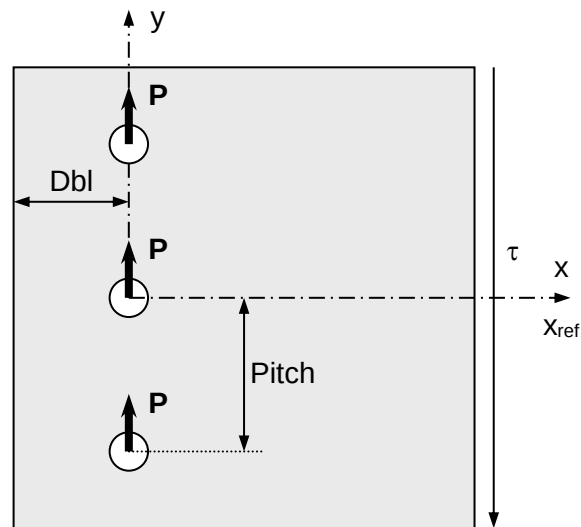
The data corresponding to this problem are the following:

- Geometry:
 - Hole diameter:5 mm.
 - Panel thickness:2 mm.
 - Fastener pitch:20 mm.
 - Distance from holes to the near edge:10 mm.
- Stress state:
 - Shear stress: $\tau = 100 \text{ N/mm}^2$.

The load transferred through the fasteners must be calculated by the user applying the equilibrium of forces, resulting:

$$P = \tau \cdot \text{Pitch} \cdot t = 4000 \text{ N}$$

This case can be associated to a single basic case, as follows:



$$\begin{aligned} \tau &= 100 \text{ N/mm}^2 & \theta &= 0^\circ \\ Dbl &= 10 \text{ mm} & \text{Pitch} &= 20 \text{ mm} \\ P &= 4000 \text{ N} \end{aligned}$$

Figure 15. Problem 4 – basic case.

Since there is only one basic case, only the stress distribution around the hole for the resultant case will be requested.

For Kt calculation purposes, the tensile principal stress of the panel, which is equal than the shear stress, will be taken as reference stress, therefore $\sigma_{ref} = \sigma_1 = \tau = 100 \text{ N/mm}^2$.

The input file containing the above data for the four problems as well as the output results file are provided on next pages.

Input data file:

```

1 Program AGUJERO. First run.
  Unloaded hole in infinite sheet.
  Biaxial far field stress plus shear.
  Stresses: Normal-X = 100 Normal-Y = 60 Shear-XY = 15.
2 General data: D      t      Sref
                  5.0    2.0    105.0
3 By-pass normal stress: n      Sigma      Ang.      Db1      Er/Ep
                        2      100.0      90.0      1000.0  0.0
                        60.0      0.0      1000.0  0.0
4 By-pass shear stress:  n      Tau        Ang.      Db1      Er/Ep
                        1      15.0      0.0      1000.0  0.0
7 Print control: Partial-Distr  Total-Distr.
                  1              1
8 Run the program.
1 Program AGUJERO. Second run.
  Unloaded hole in infinite sheet.
  Biaxial far field stress without shear.
  Stresses: Normal-X = 105 Normal-Y = 55.
3 By-pass normal stress: n      Sigma      Ang.      Db1      Er/Ep
                        2      105.0      108.435  1000.0  0.0
                        55.0      18.435  1000.0  0.0
4 By-pass shear stress:  n      Tau        Ang.      Db1      Er/Ep
                        0
7 Print control: Partial-Distr  Total-Distr.
                  0              1
8 Run the program.
1 Program AGUJERO. Third run.
  Loaded hole in a finite width sheet.
  Axial load transfer.
  Fastener load: 4000.
2 General data: D      t      Sref
                  5.0    2.0    100.0
3 By-pass normal stress: n      Sigma      Ang.      Db1      Er/Ep
                        0
4 By-pass shear stress:  n      Tau        Ang.      Db1      Er/Ep
                        0
5 Axial load transfer:  n      Load      Ang.      Db1      Dbs
                        1      4000.0    0.0      10.0    17.5
7 Print control: Partial-Distr  Total-Distr.
                  0              1
8 Run the program.
1 Program AGUJERO. Fourth run.
  Loaded hole in a finite width sheet.
  Shear load transfer.
  Fastener load: 4000.
5 Axial load transfer:  n      Load      Ang.      Db1      Dbs
                        0
6 Shear load transfer:  n      Load      Ang.      Db1      Pitch
                        1      4000.0    0.0      10.0    20.0
7 Print control: Partial-Distr  Total-Distr.
                  0              1
8 Run the program.

```

Output results file:

Program AGUJERO. First run.
 Unloaded hole in infinite sheet.
 Biaxial far field stress plus shear.
 Stresses: Normal-X = 100 Normal-Y = 60 Shear-XY = 15.

PAG: 1

```
*****
*
*   <<< I N P U T   D A T A >>>   *
*
*****
```

Execution data:
 =====

Date: 29-Apr-2021
 Agujero 4.0. (c) Airbus Defence and Space. May-2021

General data:
 =====

Hole diameter: 5.000
 Thickness: 2.000
 Reference stress: 105.0

By-pass axial stress states:
 =====

Case	Stress	Angle	Dist. to edge	Y.Mod. ratio (Er/Ep)
1	100.0	90.00	1000.000	0.0000
2	60.0	0.00	1000.000	0.0000

By-pass shear stress states:
 =====

Case	Stress	Angle	Dist. to edge	Y.Mod. ratio (Er/Ep)
1	15.0	0.00	1000.000	0.0000

Axial load transfer states:
 =====

None.

Shear load transfer states:
 =====

None.

Output results file: (cont.)

Program AGUJERO. First run.
 Unloaded hole in infinite sheet.
 Biaxial far field stress plus shear.
 Stresses: Normal-X = 100 Normal-Y = 60 Shear-XY = 15.

PAG: 2

```
*****
*
*   <<< A N A L Y S I S   R E S U L T S >>>   *
*
*****
```

Stress distribution around the hole due to BY-PASS AXIAL LOAD:
 =====

By-pass axial stress: 100.0
 Angle between reference and absolute coordinate axes: 90.00

Angle	Stress	Angle	Stress	Angle	Stress	Angle	Stress
-----	-----	-----	-----	-----	-----	-----	-----
0	-99.6	90	297.8	180	-99.6	270	297.8
5	-96.4	95	294.4	185	-96.4	275	294.4
10	-87.5	100	286.1	190	-87.5	280	286.1
15	-72.8	105	271.6	195	-72.8	285	271.6
20	-52.6	110	251.2	200	-52.6	290	251.2
25	-27.9	115	227.3	205	-27.9	295	227.3
30	0.8	120	198.7	210	0.8	300	198.7
35	32.4	125	166.9	215	32.4	305	166.9
40	66.3	130	133.3	220	66.3	310	133.3
45	99.9	135	99.9	225	99.9	315	99.9
50	133.3	140	66.3	230	133.3	320	66.3
55	166.9	145	32.4	235	166.9	325	32.4
60	198.7	150	0.8	240	198.7	330	0.8
65	227.3	155	-27.9	245	227.3	335	-27.9
70	251.2	160	-52.6	250	251.2	340	-52.6
75	271.6	165	-72.8	255	271.6	345	-72.8
80	286.1	170	-87.5	260	286.1	350	-87.5
85	294.4	175	-96.4	265	294.4	355	-96.4

Output results file: (cont.)

Program AGUJERO. First run.
 Unloaded hole in infinite sheet.
 Biaxial far field stress plus shear.
 Stresses: Normal-X = 100 Normal-Y = 60 Shear-XY = 15.

PAG: 3

```
*****
*
*   <<< A N A L Y S I S   R E S U L T S >>>
*
*****
```

Stress distribution around the hole due to BY-PASS AXIAL LOAD:
 =====

By-pass axial stress: 60.0
 Angle between reference and absolute coordinate axes: 0.00

Angle	Stress	Angle	Stress	Angle	Stress	Angle	Stress
-----	-----	-----	-----	-----	-----	-----	-----
0	178.7	90	-59.8	180	178.7	270	-59.8
5	176.6	95	-57.8	185	176.6	275	-57.8
10	171.6	100	-52.5	190	171.6	280	-52.5
15	163.0	105	-43.7	195	163.0	285	-43.7
20	150.7	110	-31.5	200	150.7	290	-31.5
25	136.4	115	-16.8	205	136.4	295	-16.8
30	119.2	120	0.5	210	119.2	300	0.5
35	100.2	125	19.4	215	100.2	305	19.4
40	80.0	130	39.8	220	80.0	310	39.8
45	59.9	135	59.9	225	59.9	315	59.9
50	39.8	140	80.0	230	39.8	320	80.0
55	19.4	145	100.2	235	19.4	325	100.2
60	0.5	150	119.2	240	0.5	330	119.2
65	-16.8	155	136.4	245	-16.8	335	136.4
70	-31.5	160	150.7	250	-31.5	340	150.7
75	-43.7	165	163.0	255	-43.7	345	163.0
80	-52.5	170	171.6	260	-52.5	350	171.6
85	-57.8	175	176.6	265	-57.8	355	176.6

Output results file: (cont.)

Program AGUJERO. First run.
 Unloaded hole in infinite sheet.
 Biaxial far field stress plus shear.
 Stresses: Normal-X = 100 Normal-Y = 60 Shear-XY = 15.

PAG: 4

```
*****
*
*   <<< A N A L Y S I S   R E S U L T S >>>   *
*
*****
```

Stress distribution around the hole due to BY-PASS SHEAR LOAD:
 =====

By-pass shear stress: 15.0
 Angle between reference and absolute coordinate axes: 0.00

Angle	Stress	Angle	Stress	Angle	Stress	Angle	Stress
-----	-----	-----	-----	-----	-----	-----	-----
0	0.0	90	0.0	180	0.0	270	0.0
5	-10.3	95	10.4	185	-10.3	275	10.4
10	-20.4	100	20.5	190	-20.4	280	20.5
15	-29.8	105	30.0	195	-29.8	285	30.0
20	-38.3	110	38.5	200	-38.3	290	38.5
25	-45.7	115	46.0	205	-45.7	295	46.0
30	-51.7	120	51.9	210	-51.7	300	51.9
35	-56.0	125	56.1	215	-56.0	305	56.1
40	-58.6	130	58.7	220	-58.6	310	58.7
45	-59.5	135	59.5	225	-59.5	315	59.5
50	-58.7	140	58.6	230	-58.7	320	58.6
55	-56.1	145	56.0	235	-56.1	325	56.0
60	-51.9	150	51.7	240	-51.9	330	51.7
65	-46.0	155	45.7	245	-46.0	335	45.7
70	-38.5	160	38.3	250	-38.5	340	38.3
75	-30.0	165	29.8	255	-30.0	345	29.8
80	-20.5	170	20.4	260	-20.5	350	20.4
85	-10.4	175	10.3	265	-10.4	355	10.3

Output results file: (cont.)

Program AGUJERO. First run.
 Unloaded hole in infinite sheet.
 Biaxial far field stress plus shear.
 Stresses: Normal-X = 100 Normal-Y = 60 Shear-XY = 15.

PAG: 5

```
*****
*
*   <<< A N A L Y S I S   R E S U L T S >>>   *
*
*****
```

Total stress distribution around hole:
 =====

Angle	Stress	Angle	Stress	Angle	Stress	Angle	Stress
0	79.1	90	238.0	180	79.1	270	238.0
5	69.9	95	247.0	185	69.9	275	247.0
10	63.7	100	254.1	190	63.7	280	254.1
15	60.4	105	257.9	195	60.4	285	257.9
20	59.9	110	258.3	200	59.9	290	258.3
25	62.7	115	256.5	205	62.7	295	256.5
30	68.3	120	251.0	210	68.3	300	251.0
35	76.5	125	242.5	215	76.5	305	242.5
40	87.6	130	231.8	220	87.6	310	231.8
45	100.3	135	219.4	225	100.3	315	219.4
50	114.4	140	204.9	230	114.4	320	204.9
55	130.2	145	188.6	235	130.2	325	188.6
60	147.3	150	171.7	240	147.3	330	171.7
65	164.6	155	154.2	245	164.6	335	154.2
70	181.2	160	136.5	250	181.2	340	136.5
75	197.9	165	120.0	255	197.9	345	120.0
80	213.0	170	104.5	260	213.0	350	104.5
85	226.1	175	90.6	265	226.1	355	90.6

Stress concentration factor:
 =====

```
Applied load .....: 0.0
Load angle .....: 0.00

Maximum stress .....: 258.4
Angle of Max .....: 108.28

Reference stress .....: 105.0
Bearing stress .....: 0.0

Kt = (Maximum)/(Reference) .....: 2.4607
Kt = (Maximum)/(Bearing) .....: 0.0000
```

Output results file: (cont.)

Program AGUJERO. Second run.
 Unloaded hole in infinite sheet.
 Biaxial far field stress without shear.
 Stresses: Normal-X = 105 Normal-Y = 55.

PAG: 6

```
*****
*
*   <<< I N P U T   D A T A >>>   *
*
*****
```

By-pass axial stress states:
 =====

Case	Stress	Angle	Dist. to edge	Y.Mod. ratio (Er/Ep)
----	-----	-----	-----	-----
1	105.0	108.43	1000.000	0.0000
2	55.0	18.43	1000.000	0.0000

By-pass shear stress states:
 =====

None.

Axial load transfer states:
 =====

None.

Shear load transfer states:
 =====

None.

Output results file: (cont.)

Program AGUJERO. Second run.
 Unloaded hole in infinite sheet.
 Biaxial far field stress without shear.
 Stresses: Normal-X = 105 Normal-Y = 55.

PAG: 7

```
*****
*
*   <<< A N A L Y S I S   R E S U L T S >>>   *
*
*****
```

Total stress distribution around hole:

=====

Angle	Stress	Angle	Stress	Angle	Stress	Angle	Stress
0	79.6	90	238.2	180	79.6	270	238.2
5	70.4	95	247.9	185	70.4	275	247.9
10	63.6	100	253.6	190	63.6	280	253.6
15	59.9	105	257.0	195	59.9	285	257.0
20	59.3	110	257.7	200	59.3	290	257.7
25	61.8	115	255.0	205	61.8	295	255.0
30	67.5	120	250.5	210	67.5	300	250.5
35	75.8	125	242.1	215	75.8	305	242.1
40	86.7	130	231.5	220	86.7	310	231.5
45	100.0	135	219.1	225	100.0	315	219.1
50	114.9	140	204.2	230	114.9	320	204.2
55	131.0	145	188.0	235	131.0	325	188.0
60	148.4	150	171.1	240	148.4	330	171.1
65	165.0	155	154.7	245	165.0	335	154.7
70	181.7	160	137.5	250	181.7	340	137.5
75	198.1	165	120.8	255	198.1	345	120.8
80	213.7	170	105.4	260	213.7	350	105.4
85	227.2	175	91.5	265	227.2	355	91.5

Stress concentration factor:

=====

Applied load: 0.0
 Load angle: 0.00

Maximum stress: 257.9
 Angle of Max: 108.58

Reference stress: 105.0
 Bearing stress: 0.0

Kt = (Maximum)/(Reference): 2.4558
 Kt = (Maximum)/(Bearing): 0.0000

Output results file: (cont.)

Program AGUJERO. Third run.
Loaded hole in a finite width sheet.
Axial load transfer.
Fastener load: 4000.

PAG: 8

```
*****
*                                     *
*   <<< I N P U T   D A T A >>>   *
*                                     *
*****
```

General data:
=====

```
Hole diameter .....:      5.000
Thickness .....:      2.000
Reference stress .....:     100.0
```

By-pass axial stress states:
=====

None.

By-pass shear stress states:
=====

None.

Axial load transfer states:
=====

Case	Load	Angle	Dist. to edge	Dist. to end
----	-----	-----	-----	-----
1	4000.0	0.00	10.000	17.500

Shear load transfer states:
=====

None.

Output results file: (cont.)

Program AGUJERO. Third run.
Loaded hole in a finite width sheet.
Axial load transfer.
Fastener load: 4000.

PAG: 9

```
*****
*
*   <<< A N A L Y S I S   R E S U L T S >>>   *
*
*****
```

Total stress distribution around hole:
=====

Angle	Stress	Angle	Stress	Angle	Stress	Angle	Stress
0	527.5	90	308.1	180	527.5	270	-83.2
5	528.5	95	309.6	185	495.7	275	-79.4
10	485.6	100	315.0	190	466.9	280	-68.6
15	455.5	105	321.1	195	432.1	285	-50.7
20	441.1	110	323.7	200	392.7	290	-26.1
25	423.3	115	326.0	205	351.5	295	4.3
30	410.7	120	337.2	210	306.7	300	39.9
35	397.0	125	350.2	215	259.9	305	79.8
40	381.6	130	358.7	220	212.7	310	123.4
45	369.4	135	369.4	225	167.5	315	167.5
50	358.7	140	381.6	230	123.4	320	212.7
55	350.2	145	397.0	235	79.8	325	259.9
60	337.2	150	410.7	240	39.9	330	306.7
65	326.0	155	423.3	245	4.3	335	351.5
70	323.7	160	441.1	250	-26.1	340	392.7
75	321.1	165	455.5	255	-50.7	345	432.1
80	315.0	170	485.6	260	-68.6	350	466.9
85	309.6	175	528.5	265	-79.4	355	495.7

Stress concentration factor:
=====

```
Applied load .....: 4000.0
Load angle .....: 90.00

Maximum stress .....: 533.5
Angle of Max .....: 2.61

Reference stress .....: 100.0
Bearing stress .....: 400.0

Kt = (Maximum)/(Reference) .....: 5.3352
Kt = (Maximum)/(Bearing) .....: 1.3338
```

Output results file: (cont.)

Program AGUJERO. Fourth run.
Loaded hole in a finite width sheet.
Shear load transfer.
Fastener load: 4000.

PAG: 10

```
*****
*                                     *
*   <<< I N P U T   D A T A >>>   *
*                                     *
*****
```

By-pass axial stress states:
=====

None.

By-pass shear stress states:
=====

None.

Axial load transfer states:
=====

None.

Shear load transfer states:
=====

Case	Load	Angle	Dist. to edge	Fast. pitch
----	-----	-----	-----	-----
1	4000.0	0.00	10.000	20.000

Output results file: (cont.)

Program AGUJERO. Fourth run.
Loaded hole in a finite width sheet.
Shear load transfer.
Fastener load: 4000.

PAG: 11

```
*****
*
*   <<< A N A L Y S I S   R E S U L T S >>>   *
*
*****
```

Total stress distribution around hole:
=====

Angle	Stress	Angle	Stress	Angle	Stress	Angle	Stress
0	319.4	90	369.1	180	331.7	270	42.0
5	349.8	95	328.3	185	353.2	275	2.5
10	386.0	100	289.9	190	355.5	280	-35.6
15	433.1	105	253.8	195	357.9	285	-71.1
20	476.1	110	220.8	200	357.3	290	-102.5
25	516.4	115	192.7	205	355.6	295	-128.3
30	549.4	120	169.4	210	350.0	300	-146.8
35	574.4	125	151.4	215	341.2	305	-156.8
40	592.9	130	139.7	220	329.9	310	-157.6
45	602.8	135	133.6	225	316.0	315	-149.0
50	601.7	140	131.7	230	297.8	320	-130.1
55	592.2	145	135.3	235	275.9	325	-100.6
60	576.5	150	145.1	240	251.4	330	-61.1
65	553.2	155	160.3	245	223.3	335	-11.8
70	522.8	160	180.1	250	191.5	340	45.9
75	488.7	165	206.0	255	157.2	345	111.6
80	450.6	170	235.0	260	120.3	350	182.3
85	409.9	175	279.5	265	81.6	355	258.8

Stress concentration factor:
=====

```
Applied load .....: 4000.0
Load angle .....: 90.00

Maximum stress .....: 603.7
Angle of Max .....: 47.01

Reference stress .....: 100.0
Bearing stress .....: 400.0

Kt = (Maximum)/(Reference) .....: 6.0365
Kt = (Maximum)/(Bearing) .....: 1.5091
```

